Mostafa Abdelmeguid Ramadan

Computer Science Student | AI Developer | Python Developer

Alexandria, Egypt | Matsoo27@gmail.com | github.com/MatsoDev

PROFILE

Computer Science student on the Cybersecurity track at Pharos University in Alexandria, building and shipping independent projects outside coursework. Work spans YOLOv8 object detection pipelines, multithreaded Python automation, FastAPI backends, and ESP32-CAM edge prototypes. Currently focused on AI agent architectures and Model Context Protocol (MCP) tooling for practical developer workflows.

CORE COMPETENCIES

Computer Vision and Object Detection; AI Agent Development and MCP; Python Automation and Scripting; Machine Learning Pipeline Design; Backend API Development; Multithreaded Systems; Embedded AI and Edge Computing; Prompt Engineering; REST API Design and Integration.

SELECTED PROJECTS

AI Automation Tools. Python automation suite for web scraping and multi-source data processing, with separated scraping, pipeline, and GUI layers and concurrent threading to keep the Tkinter interface responsive.

Computer Vision Pipeline. YOLOv8 and OpenCV object detection from dataset preparation and annotation through training, validation, and automation integration; evaluated accuracy and inference speed trade-offs.

ESP32-CAM Edge Prototype. Live video streaming and user authentication on embedded hardware within microcontroller memory, compute, and connectivity constraints.

MCP and AI Agent Research. Studied Model Context Protocol architecture and multi-agent coordination patterns for AI-assisted developer tooling.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, HTML, CSS. **AI/ML:** OpenCV, YOLOv8, Machine Learning, AI Agents, MCP. **Backend:** FastAPI, REST APIs, SQLAlchemy, JWT. **Automation:** Selenium, Web Scraping, Multithreading, Tkinter. **Embedded:** ESP32-CAM, Arduino, Embedded Networking. **Tools:** Git, GitHub, Docker, Linux/WSL, VS Code, Figma.

EDUCATION & CERTIFICATION

BSc Computer Science, Cybersecurity specialization - Pharos University in Alexandria; expected 2027. Coursework: AI, Machine Learning, Pattern Recognition, Digital Forensics, Advanced Computer Networks, and Web Security.

UI/UX Design Training (100 hours) - Arabian Academy in Alexandria, 2025. Figma wireframes and prototypes, design systems, and usability evaluation.

LANGUAGES

Arabic (Native) | English (Upper-Intermediate, B2)